Introduction to MeshCore

A beginner's guide to off-grid text messaging

Clemson Amateur Radio Club · WD4EOG

MESHCORE

START HERE

What's in this booklet

How to use this guide

You don't need any radio experience. Work through the pages in order and you'll go from an unboxed radio to sending your first message.

Radio words appear in **bold orange** the first time they show up, with a plain-English explanation right there. Every one of them is also in the **Glossary** on pages 24–25.

1 **Meet MeshCore** — What it is, radio basics, mesh networks, node types — p. 3–6

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What is MeshCore?

MeshCore lets you send **text messages with small, low-cost radios** instead of cell towers or the internet.

Each radio is called a **node**. Your phone talks to your own node over Bluetooth, and the node sends your message through the air to other nodes, either directly or by passing it along through other nodes on the way.

Because it doesn't need cell service, Wi-Fi, or satellites, it keeps working when those are down.

Great for

- Emergencies and disaster recovery
- Hiking, camping, and outdoor events
- Club activities, public service, and learning radio



Works off-grid

No cell signal, Wi-Fi, or internet needed. The radios talk to each other directly.



Private messages

Direct messages are **encrypted**: scrambled so only the person you sent them to can read them.



Long range

Messages travel miles, not feet. Nodes placed up high (rooftops, hills) reach much farther.



Free and open source

The radio software is free under the MIT license. Free apps for Android, iPhone, Windows, Mac, and web.

Radio basics in one page



LoRa: the radio inside

LoRa (short for “Long Range”) is a radio technology built to send small bits of data very far on very little power.

The trade-off is speed. It’s perfect for short text messages, but too slow for photos, video, or voice calls.



Frequency: the “station”

Radios send signals on a **frequency**, like picking a station on an FM radio. It’s measured in **MHz** (megahertz, millions of cycles per second).

In the US, MeshCore uses the **902–928 MHz** band. Every node in an area must be tuned to the same settings to hear each other.



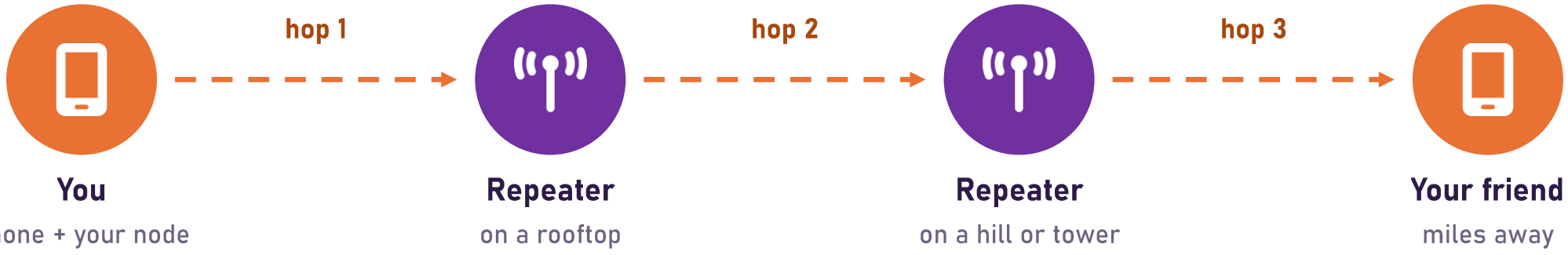
Do I need a license?

No, not at the default power settings. In the US, the 902–928 MHz band is open to everyone for low-power devices, the same band some cordless phones and smart-home gadgets use.

Buy radios made for this band and use the recommended settings. With an amateur (ham) radio license you can legally use higher power, but it’s typically not needed.

How a mesh network works

In a **mesh** network, a message can jump from node to node to reach someone too far away to hear you directly. Each jump is called a **hop**. Special nodes called **repeaters**, usually mounted up high, do the passing-along.



Only repeaters relay

Your personal node never re-sends other people's messages. Fewer radios talking at once means fewer messages bump into each other and get lost. This is a key way MeshCore differs from other mesh systems.

Height beats power

LoRa signals travel in fairly straight lines and are blocked by hills and buildings. A repeater up high can "see" a huge area. A message can make up to 64 hops.

Three jobs a node can do

The same radio hardware can do different jobs depending on its **firmware**: the software that runs inside the radio itself (separate from the app on your phone). You pick the job when you install the firmware.

Companion

Everyone needs one

Your personal radio. It pairs with the MeshCore app on your phone or computer and sends and receives your messages.

It does **not** relay messages for others.

Think of it as: your own walkie-talkie for texting.

Repeater

Set up by clubs and hosts

Extends the network's reach. It sits somewhere high, often solar powered, and passes messages along toward where they're going.

You don't chat on it, and phones can't connect to it over Bluetooth.

Think of it as: a relay tower.

Room Server

Optional

A shared message board. It stores posts so people who were out of range can log in later and catch up (the 32 most recent unseen messages).

Best run on its own device, separate from a repeater.

Think of it as: a bulletin board or email inbox.

No phone? Some devices, like the LilyGO T-Deck, have their own screen and keyboard and run special **standalone** firmware, so they work all by themselves.

What you need

A companion node

One per person who wants to send messages. See the next page for good choices.

The MeshCore app

Free for Android, iPhone, Windows, Mac, Linux, or in a web browser.

A computer + USB data cable

Used once to install the firmware. Needs Google Chrome or Microsoft Edge.

Nice to have

A better antenna and a USB power bank. A charged radio is a working radio.

Buying tip: get the 915 MHz version

Many radios are sold in several versions for different countries. In the US, choose the one labeled **915 MHz** (sometimes "US915" or "902–928 MHz").

433 MHz and 868 MHz versions are for other parts of the world and **won't work** with our local mesh.

Also make sure the antenna is included, and never power on a radio without its antenna attached; it can damage the radio.

Picking a radio

Companion nodes

Pair with your phone. Pick one of these to start.

- **Seeed Wio Tracker L1 Pro**
- **Elecrow ThinkNode M1**
- **RAK WisMesh Pocket**
- **RAK WisMesh Tag**
  rugged, weatherproof
- **Heltec V3**
  budget board; usually needs a case and battery; short Bluetooth range

Standalone devices

Built-in screen and keyboard. No phone needed.

- **LilyGO T-Deck Plus**
- **LilyGO T-Deck Pro**
- **LilyGO T-LoRa Pager**
- **LilyGO T-Watch S3 Plus**
  a wristwatch

Repeater

For extending range from a high spot.

- **Seeed SenseCAP Solar Node P1 Pro**
  solar powered, made for outdoors

These are the devices recommended on meshcore.io. Many more work; the full list is at **flasher.meshcore.io**. Many radios ship with different firmware (often Meshtastic); you'll replace it with MeshCore in step 2.

Install the MeshCore app

On a phone

- Open the **Google Play Store** or **Apple App Store**.
- Search for **MeshCore** by **Liam Cottle** and install it.

On a computer

- Download the app for **Windows**, **Mac**, or **Linux** from **meshcore.io**.
- The computer versions are fully free and unlocked.

In a web browser

- Go to **app.meshcore.nz** in Chrome or Edge.
- Handy for trying things out or using a radio plugged in by USB.

Save an offline copy

When the internet is down is exactly when you'll want MeshCore. Install the app (and keep the installer) before you need it.

Is it free?

Yes, for messaging. The phone app has an optional in-app purchase that removes a wait timer for managing repeaters remotely. You don't need it to chat.

Load the MeshCore firmware



What's going on here?

Firmware is the software that runs inside the radio. **Flashing** means copying new firmware onto it.

You only do this once to set up, and again later if you want updates.

DFU mode (Device Firmware Update) is a special mode where the radio waits to receive new firmware.

1. On a computer, open **flasher.meshcore.io** in **Google Chrome** or **Microsoft Edge**. (Other browsers can't talk to USB devices.)
2. Plug your radio into the computer with a USB cable. If nothing shows up later, try another cable; some only charge.
3. Find your device in the list and click it.
4. Choose **Companion Bluetooth** firmware. (Not "Companion USB": that version won't show up on your phone.)
5. If the flasher shows an **Enter DFU mode** button, click it and pick your radio from the pop-up.
6. Click **Flash!**, pick your radio again, and wait until it says it's done. Don't unplug it partway.

Switching from other firmware like Meshtastic, or something acting up? Click **Erase Flash** first, then **Flash!**

Connect your phone to your node

1

Turn on your node

Power it up. If it has a screen, you'll see its name and a PIN.

2

Tap Connect

Open the MeshCore app and tap **Connect** in the top-right corner.

3

Pick your node

Tap the **MeshCore-XXXX** entry whose name matches your node's screen.

4

Enter the PIN

Type the PIN shown on the node. **No screen? The PIN is 123456.**

That's it! You'll land on the main screen, connected and ready.

Bluetooth is only the short hop between your phone and your node (a few feet to across a room). The long-distance part is your node's LoRa radio.

Won't pair? Restart your phone and try again. Still missing? Check that you flashed **Companion Bluetooth**, not Companion USB (see page 23).

Set your name and radio settings

A. Pick your display name

This is the name other people see on the mesh. A first name, nickname, or your ham call sign works well. Keep it short.

1. Tap the **settings** (gear) icon, top right.
2. Tap the name field, clear it, and type your name.
3. Tap the **check mark** (top right) to save.

B. Choose the USA preset

A **preset** is a ready-made bundle of radio settings. Picking the right one makes sure you're on the same "station" as everyone else, and legal.

1. In **settings**, find the radio settings section.
2. Tap **Choose Preset**, then tap **USA**.
3. Tap the **check mark** to save.

USA preset **What you should see**	**910.525 MHz** frequency	**BW 62.5** bandwidth	**SF 7** spreading factor	**CR 5** coding rate

Every node must match exactly. Our local mesh uses the standard USA preset. These numbers are explained on the next page.

What those radio numbers mean

910.525 MHz

Frequency

The “station” everyone tunes to. If your frequency is different, you simply won’t hear each other.

62.5 kHz

Bandwidth (BW)

How wide a slice of the radio dial each signal uses. A narrow slice picks up less background noise and squeezes between interference.

SF 7

Spreading factor (SF)

How stretched-out each message is. Higher numbers reach farther but take longer to send and tie up the airwaves more.

CR 5

Coding rate (CR)

How much extra backup data is added so a garbled message can be repaired. 5 adds the least; 7 or 8 help on weak links.



An easy way to picture it

Imagine two people talking in a noisy room. Far apart, they speak slowly and clearly (a high spreading factor). Close together, they can talk faster (a low one). It’s always a balance between speed and getting through the noise. You don’t need to change any of these yourself; the preset handles it.

Say hello on the Public channel

A **channel** is a group chat. Everyone starts with the **Public** channel, a big open group chat for everyone on the mesh. It's the easiest way to test that your radio works.

1. Tap the **Channels** tab at the bottom of the screen.
2. Tap the **Public** channel.
3. Type a message, like "Hello from Clemson!", and tap **send**.

What you'll see

- **"Heard 1 repeat"** means a nearby repeater picked up your message and passed it on. Great sign!
- **"Sent"** means it went out, but your node didn't hear a repeater pass it along. You may be out of range.
- There's no way to know exactly who received a channel message.

Public means public

Anyone on the mesh can read the Public channel. Never post private information there; use a private message (page 16) or a private channel instead.

Add contacts

To send someone a private message, **you both have to add each other as contacts**. If they haven't added you, your messages to them won't get through. There are two ways to swap contact info:

In person: scan a QR code

Private. Best option when you're together.

To show your code: tap the ⋮ (three dots) menu, top right → **My Contact Code**.

To scan theirs: ⋮ menu → **Add Contact** → **Scan QR Code**, then point your camera at it. (Allow camera access if asked.)

Then swap: they scan yours too. The code holds your name and your **public key**: a code others use to lock messages so only your radio can unlock them.

Over the air: send an advert

Public. Anyone in range gets your contact info.

An **advert** is a short broadcast that announces your name and public key so others can add you. Tap the **signal icon** (top right) and pick one:

- **Advert (Zero Hop):** only radios close enough to hear you directly get it. A **zero hop** message is never passed along by repeaters.
- **Advert (Flood Routed):** every repeater that hears it passes it on, so it can reach the whole mesh. That's called **flooding**.

Send a private message

Private messages (also called **direct messages**) are encrypted: only the person you send them to can read them, even though the radio signal goes everywhere.

1. Tap the contact you want to message.
2. Type your message and tap **send**.
3. Watch the status under your message:

Delivered: your friend's radio confirmed it got the message.

Failed (shown in red): the app tried several times but never got a confirmation back.

Why might it fail?

- You're out of range, with no repeaters in between.
- Their radio is turned off.
- They haven't added you as a contact.
- Radio interference.
- Sometimes it arrived, but the confirmation didn't make it back.

To try again

Press and hold the red message, then tap **Retry Sending**. Once you see **Delivered**, you know for sure it arrived.

Behind the scenes: how messages find the way

You don't have to do anything here; MeshCore handles it. But knowing how it works explains a lot about what you see.

1. The first message floods

The first time you message someone, every repeater that hears it passes it along once, spreading it in all directions until it finds them.

2. A path comes back

Their radio sends back a delivery report listing which repeaters the message went through. That list is called the **path**, and your app saves it.

3. Next time, it goes direct

Later messages carry the path. Only the repeaters on that path pass it on; the others stay quiet, keeping the airwaves clear for everyone.

If the path breaks

Repeaters go offline and people move. If a direct message fails three times, the app floods it again to discover a fresh path.

Channels always flood

Group messages (like Public) have no single destination, so they're always flooded through every repeater that hears them.

Channels and room servers

Public channel

One big open chat for everyone on the mesh. Already set up for you.

Private channels

A group chat locked with a secret key. Only people you share it with (for example by QR code) can read it. Great for family or a team.

Hashtag channels

Open topic chats anyone can join by typing the name, like **#clemson**. Treat them as public.

Channels are live: if you're out of range when a message is sent, you miss it.

Room servers

A **room server** works like an email inbox or bulletin board: it **stores** messages so you can catch up later.

- Room servers show up in your contacts like a person. Tap one and log in with its password.
- The usual guest password is `hello` unless the owner changed it.
- When you log in, you get up to **32** messages you haven't seen yet.

Get the best range

Antenna straight up

Keep the antenna vertical, pointing at the sky. Lying flat in a pocket can cut your range a lot.

Go high, go outside

Higher is better. Signals work best outdoors and struggle inside buildings, cars, and basements.

Line of sight

Line of sight means a clear, straight path between two antennas. Hills, buildings, trees, even people block LoRa.

Upgrade the antenna

The little antenna that comes in the box is often the weakest part. A better 915 MHz antenna is a cheap upgrade.

Stay charged

A dead radio can't help you. Most can be used while plugged into a charger or power bank.

Back up your settings

In app settings, use **Export Settings** now and then. It saves your identity and contacts.

Check your signal with a ping

A **ping** is a quick test: your node calls a nearby repeater and the repeater answers, reporting how well each heard the other.

1. In your contacts, find a repeater and tap the ⋮ (three dots) next to it.
2. Tap **Ping (Zero Hop)**.
3. **Green pop-up:** success, with signal numbers. **Red "Timeout":** the two radios couldn't hear each other directly.

A ping only goes **one hop**: you must be within range of that specific repeater. It's a great way to find the best spot to stand or mount an antenna.

RSSI: how loud the signal is

RSSI (received signal strength). These numbers are negative; closer to zero is stronger.



weak (about -120) **strong (about -60)**

Think of it as the volume of the other radio's voice.

SNR: how clear the signal is

SNR (signal-to-noise ratio) compares the signal to the background static. Higher is better.



barely there (below 0) **crystal clear (+10 or more)**

LoRa can still work slightly below zero, which is part of its magic.

Our local mesh

Two MeshCore repeaters cover the Clemson area from the roof of **Byrnes Hall**. If you can reach one of them, your messages can travel much farther than your own radio could manage alone.

WD4EOG repeater

Run by the Clemson Amateur Radio Club
Location: Byrnes Hall rooftop

Anderson Radio Club repeater

Run by the Anderson Radio Club
Location: Byrnes Hall rooftop

Find nodes on the map

See repeaters near you at **map.meshcore.io**. Your app can also show who it has heard.

Add yourself (optional)

In the app: ⋮ menu → **Internet Map** → ⋮ → **Add me to the Map**. This shares your location publicly.

Get help from the club

Bring your radio to a club meeting. Members are happy to help you flash, pair, and test it.

Setting up a repeater (for hosts)

1. **Flash it.** On flasher.meshcore.io, choose your device and pick **Repeater** firmware this time.

2. **Configure it over USB.** Keep it plugged in and open **config.meshcore.io** (or the flasher's **Console**). Set a name, the USA radio settings, and optionally its location.

3. **Change the admin password!** Every repeater starts with the password `password`. Change it right away.

4. **Mount it high** with a good antenna and steady power. Solar plus a battery is common for rooftops and towers.

5. **Manage it remotely.** In the app, log in to the repeater with its admin password over the radio to check stats or change settings.

*Replace the parts in < > with your own values. Full command list: **docs.meshcore.io/cli_commands**. Run repeaters and room servers on separate devices.*

Handy console commands

```
set name <repeater name>
```
name it

```
set radio 910.525,62.5,7,5
```
USA preset, then reboot

```
password <new password>
```
new admin password

```
set lat <latitude>
```
decimal degrees, e.g. from Google Maps

```
set lon <longitude>
```
decimal degrees (west is negative)

```
reboot
```
restart to apply

Troubleshooting

If this happens…	Try this
My node doesn't show up in the app's Bluetooth list.	Make sure you flashed **Companion Bluetooth**, not Companion USB. Restart the node and your phone.
It's asking for a PIN.	Use the PIN on the node's screen. No screen? Try **123456**.
Bluetooth keeps disconnecting (Heltec V3).	This board's Bluetooth antenna is tiny. Keep your phone close to it.
I send messages but nobody hears me.	Check you chose the **USA** preset, the antenna is on and pointing up, and you're outside or up high. Ping a repeater.
My private message says Failed.	Both of you must add each other. Their radio must be on and in range. Press and hold → **Retry Sending**.
Contacts are missing or "last seen" many days ago.	A device's clock is probably wrong. Radios need the right time; ask the repeater owner to sync its clock.
Everything is acting strange.	App: **Export Settings** → **Factory Reset** → **Import Settings**. Or reflash: **Erase Flash** then **Flash!**

Glossary

Advert	A broadcast announcing your name and public key so others can add you.
Bandwidth (BW)	How wide a slice of the radio dial a signal uses, measured in kHz.
Bluetooth (BLE)	The short-range wireless link between your phone and your node.
Channel	A group chat: public, private (secret key), or hashtag (join by name).
Coding rate (CR)	Extra backup data added to each message so small errors can be fixed.
Companion	Your personal node. Pairs with the app; doesn't relay others' messages.
DFU mode	Device Firmware Update mode: the radio waits to receive new firmware.
Direct message	A private, encrypted message to one contact.
Encryption	Scrambling a message so only the intended person can read it.
Firmware	The software that runs inside the radio itself.
Flashing	Installing firmware onto a radio.
Flood	Sending a message so every repeater that hears it passes it on once.
Frequency / MHz	The "station" a radio uses, in megahertz (millions of cycles per second).
Hop	One jump from one node to the next. Messages can make up to 64 hops.

Glossary (continued)

Line of sight	A clear, straight path between two antennas with nothing in the way.
LoRa	“Long Range”: radio technology that sends small data far on little power.
Mesh	A network where messages are passed from node to node to reach farther.
Node	Any MeshCore radio: companion, repeater, or room server.
Path	The list of repeaters a message took, saved so later messages go direct.
Ping	A quick one-hop test of the signal between your node and a repeater.
Preset	A ready-made bundle of radio settings for your region (ours: USA).
Public key	A code you share so others can encrypt messages only your radio can read.
Repeater	A node, usually mounted high, that passes messages along to extend range.
Room server	A node that stores messages so people can log in and catch up later.
RSSI / SNR	Signal strength (nearer 0 is stronger) and clarity over noise (higher is better).
Spreading factor (SF)	How stretched-out a signal is. Higher reaches farther but is slower.
Standalone	A device with its own screen and keyboard that works without a phone.
Zero hop	Sent only to radios that hear you directly; repeaters don’t pass it on.

Links and resources

meshcore.io
Official site and app downloads

flasher.meshcore.io
Install or update firmware

app.meshcore.nz
Web version of the app

config.meshcore.io
Set up repeaters over USB

docs.meshcore.io
Official docs and FAQ

map.meshcore.io
Map of MeshCore nodes

meshcore.gg
MeshCore community Discord

MeshCore Quick Start Guide
By Liam Cottle, on meshcore.io

wd4eog.github.io
Clemson Amateur Radio Club

73, and happy meshing!

"73" is ham radio shorthand for "best regards." Questions? Find us at a club meeting or at wd4eog.github.io.

MESHCORE